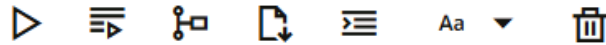


[SQL Worksheet]*



```
170
171 --Control Structure: Scenario-1
172
173 DECLARE
174     CURSOR c_customers IS
175     SELECT CustomerId,Name,DOB
176     FROM CUSTOMERS;
177     v_age NUMBER;
178 BEGIN
179     FOR cust_rec IN c_customers LOOP
180         v_age:=trunc(months_between(SYSDATE,cust_rec.DOB)/12);
181         IF v_age>60 THEN
182             UPDATE LOANS
183             SET INTERESTRATE=INTERESTRATE-(INTERESTRATE*0.1)
184             where CustomerID=cust_rec.CustomerID;
```

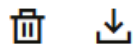
Query result

Script output

DBMS output

Explain Plan

SQL history



```
CURSOR c_customers IS
  SELECT CustomerId,Name,DOB
  FROM CUSTOMERS;...
```

[Show more...](#)

Rahul Sharma(Age: 68)-1% discount is applied
Amit Verma(Age: 66)-1% discount is applied
Kiran Kumar(Age: 70)-1% discount is applied
Neha Joshi(Age: 63)-1% discount is applied

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.011



FreeSQL

>_ Worksheet



SQL Tutorials

[SQL Worksheet]*



Aa



```
190 /
191
192 --scenario-2
193
194 SET SERVEROUTPUT ON;
195 DECLARE
196     CURSOR c_customers IS
197     SELECT CustomerID, Name, Balance
198     FROM Customers;
199 BEGIN
200     FOR cust_rec IN c_customers LOOP
201         IF cust_rec.Balance > 10000 THEN
202             UPDATE Customers
203             SET IsVIP = 'TRUE'
204             WHERE CustomerID = cust_rec.CustomerID;
205     END LOOP;
206 END;
```

Query result

Script output

DBMS output

Explain Plan

SQL history



```
SQL> DECLARE
      CURSOR c_customers IS
      SELECT CustomerID, Name, Balance
      FROM Customers;...
```

[Show more...](#)

Amit Verma (Balance: 25000) - Promoted to VIP.
Kiran Kumar (Balance: 12000) - Promoted to VIP.

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.019

[SQL Worksheet]*



Aa



```
218 --scenario-3
219
220 SET SERVEROUTPUT ON;
221 DECLARE
222     CURSOR c_due IS
223     SELECT L.LoanID, L.CustomerID, c.Name, L.EndDate, L.LoanAmount
224     FROM LOANS L
225     JOIN CUSTOMERS c ON L.CustomerID=c.CustomerID
226     WHERE L.EndDate between SYSDATE and SYSDATE+30;
227 BEGIN
228     FOR loan_rec IN c_due LOOP
229         DBMS_OUTPUT.PUT_LINE('REMINDER:Dear '||loan_rec.Name||
230         ',your loan ID:'||loan_rec.LoanID||
231         ',Amount:'||loan_rec.LoanAmount||
232         'is due on:'||TO_CHAR(loan_rec.EndDate,'DD-MON-YYYY'));
```

Query result

Script output

DBMS output

Explain Plan

SQL history



```
SELECT L.LoanID,L.CustomerID,c.Name,L.EndDate,L.LoanAmount
from LOANS L...
```

Show more...

```
REMINDER:Dear Rahul Sharma,your loan ID: 1,Amount: 50000 is due on 09-JUL-2026
REMINDER:Dear Priya Reddy,your loan ID: 2,Amount: 30000 is due on 04-JUL-2026
REMINDER:Dear Sneha Patel,your loan ID: 4,Amount: 20000 is due on 14-JUL-2026
REMINDER:Dear Anjali Singh,your loan ID: 6,Amount: 25000 is due on 29-JUN-2026
REMINDER:Dear Neha Joshi,your loan ID: 8,Amount: 15000 is due on 24-JUN-2026
```

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.017



[SQL Worksheet]*



Aa



```
289
290  --scenario-2
291
292  create or replace procedure UpdateSalary
293  (
294      ... e_id in number,
295      ... e_percent in NUMBER ..
296  )
297  is
298      ... e_exist number;
299  BEGIN
300      ... select count(*) into e_exist
301      ... from Employees
302      ... where EmployeeID=e_id;
303      ... if e_exist=0 THEN
```

Query result

Script output

DBMS output

Explain Plan

SQL history



Elapsed: 00:00:00.005

```
SQL> begin
      UpdateSalary(1,15);
      end;
```

Salary updated successfully for Employee ID: 1

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.015

[SQL Worksheet]*



Aa



```
323  --scenario-3
324
325  create or replace procedure AddNewCustomer
326  (
327      ... p_cus_id in NUMBER,
328      ... p_name in VARCHAR2,
329      ... p_DOB in DATE,
330      ... p_balance in NUMBER
331  )
332  IS
333      ... v_exist NUMBER;
334  BEGIN
335      ... SELECT count(*) into v_exist
336      ... from Customers
337      ... where CustomerID=p_cus_id;
```

Query result

Script output

DBMS output

Explain Plan

SQL history



Elapsed: 00:00:00.007

SQL> BEGIN

```
    AddNewCustomer(9,'Kiran Rao',TO_DATE('1992-04-12','YYYY-MM-DD'),5000);
end;
```

ERROR: ORA-20002: Customer ID 9 already exists.

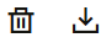
PL/SQL procedure successfully completed.

Elapsed: 00:00:00.010

[SQL Worksheet]*      Aa 

```
357
358 --Stored Procedure: Scenario-1
359
360 create or replace procedure ProcessMonthlyInterest
361 is
362 ...v_count number:=0;
363 BEGIN
364 ...update Accounts
365 ...set Balance=Balance+(Balance*0.01),LastModified=SYSDATE
366 ...WHERE AccountType='Savings';
367 ...v_count:=sql%rowcount;
368 ...COMMIT;
369 ...DBMS_OUTPUT.PUT_LINE('Monthly interest applied to-'||v_count||'-savings accounts. ');
370 exception
371 ...when others then
```

Query result Script output DBMS output Explain Plan SQL history



Elapsed: 00:00:00.007

SQL> BEGIN
 ProcessMonthlyInterest;
 end;

Monthly interest applied to 4 savings accounts.

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.012

[SQL Worksheet]* ▼



Aa ▼



```
382  --scenario-2
```

```
383
```

```
384  create or replace procedure UpdateEmployeeBonus
```

```
385  (
```

```
386    ... p_department in varchar2,
```

```
387    ... p_bonus_per in number
```

```
388  )
```

```
389  is
```

```
390    ... v_count number:=0;
```

```
391  BEGIN
```

```
392    ... update Employees
```

```
393    ... set Salary=Salary+(Salary*p_bonus_per/100)
```

```
394    ... where Department=p_department;
```

```
395    ... v_count:=sql%rowcount;
```

```
396    ... if v_count=0 then
```

Query result

Script output

DBMS output

Explain Plan

SQL history



Elapsed: 00:00:00.007

```
SQL> BEGIN
      UpdateEmployeeBonus('IT',5);
      end;
```

2 employees in IT received a 5% bonus.

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.013

[SQL Worksheet]*



Aa



```
414
415  --scenario-3
416
417  create or replace procedure TransferFunds
418  (
419    ... p_from_account IN NUMBER,
420    ... p_to_account in NUMBER,
421    ... p_amount in NUMBER
422  )
423  IS
424    ... v_from_balance number;
425    ... e_insuff_funds exception;
426  BEGIN
427    ... select Balance into v_from_balance
428    ... FROM Accounts
```

Query result

Script output

DBMS output

Explain Plan

SQL history



Elapsed: 00:00:00.021

```
SQL> begin
      TransferFunds(1,2,300);
      end;
```

Transfer successful: 300 moved from Account 1 to Account 2

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.011

[SQL Worksheet]*



Aa



```
461
462  --Functions: Scenario-1
463
464  create or replace function CalculateAge
465  (
466    ... p_DOB in DATE
467  )
468  return NUMBER
469  is
470    ... v_age NUMBER;
471  BEGIN
472    ... v_age:=trunc(months_between(SYSDATE,p_DOB)/12);
473    ... return v_age;
474  end CalculateAge;
475  /
---
```

Query result

Script output

DBMS output

Explain Plan

SQL history



```
SQL> DECLARE
      v_age NUMBER;
      BEGIN
      v_age:=CalculateAge(TO_DATE('1990-07-20', 'YYYY-MM-DD'));...
Show more...
```

Age: 35

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.006

[SQL Worksheet]*



Aa



```
485  --scenario-2
486
487  create or replace function CalculateMaonthlyInstallment
488  (
489      ... p_loan_amount in number,
490      ... p_annual_rate in number,
491      ... p_years in number
492  )
493  return NUMBER
494  IS
495      ... v_monthly_rate number;
496      ... v_num_payments number;
497      ... v_emi number;
498  BEGIN
499      ... v_monthly_rate:=(p_annual_rate/100)/12;
```

Query result

Script output

DBMS output

Explain Plan

SQL history



```
SQL> DECLARE
      v_emi NUMBER;
      BEGIN
      v_emi:=CalculateMonthlyInstallment(500000,8.5,5);...
Show more...
```

Monthly Installment: Rs.10258.27

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.008

[SQL Worksheet]*



Aa



```
520  --scenario-3
521
522  CREATE OR REPLACE FUNCTION HasSufficientBalance
523  (
524    p_account_id IN NUMBER,
525    p_amount IN NUMBER
526  )
527  RETURN VARCHAR2
528  IS
529    v_balance NUMBER;
530  BEGIN
531    SELECT Balance INTO v_balance
532    FROM Accounts
533    WHERE AccountID=p_account_id;
534    IF v_balance >= p_amount THEN
```

Query result

Script output

DBMS output

Explain Plan

SQL history



```
SQL> DECLARE
      v_result VARCHAR2(5);
      BEGIN
      v_result := HasSufficientBalance(3, 5000);...
      Show more...
```

Sufficient balance? TRUE

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.008

[SQL Worksheet]*      Aa 

```
553 --Triggers: Scenario-1
554
555 create or replace Trigger UpdateCustomerLastModified
556 before update on Customers
557 for each row
558 BEGIN
559     :new.LastModified:=SYSDATE;
560 end UpdateCustomerLastModified;
561 /
562 update Customers set Balance=1200 where CustomerID=1;
563 select CustomerID,Name,LastModified from Customers where CustomerID=1;
564
565 --scenario-2
566
567 CREATE TABLE AuditLogg (
```

Query result Script output DBMS output Explain Plan SQL history

  Download  Execution time: 0.005 seconds

	CUSTOMERID	NAME	LASTMODIFIED
1	1	Rahul Sharma	6/20/2026, 1:07:53

[SQL Worksheet]*      Aa 

```
564
565 --scenario-2
566
567 CREATE TABLE AuditLogg (
568     AuditID NUMBER GENERATED ALWAYS AS IDENTITY,
569     TransactionID NUMBER,
570     AccountID NUMBER,
571     Amount NUMBER,
572     TransactionType VARCHAR2(10),
573     LoggedOn DATE
574 );
575 CREATE OR REPLACE TRIGGER LogTransaction
576 AFTER INSERT ON Transactions
577 FOR EACH ROW
578 BEGIN
```

Query result Script output DBMS output Explain Plan SQL history

  Download  Execution time: 0.008 seconds

	AUDITID	TRANSACTION_ID	ACCOUNTID	AMOUNT	TRANSACTIONTYF	LOGGEDON
1	2	13	1	150	Deposit	6/19/2026, 9:43:57

[SQL Worksheet]*



Aa



```
586
587  --scenario-3
588
589  create or replace TRIGGER CheckTransactionRules
590  before insert on Transactions
591  for each ROW
592  DECLARE
593      v_balance number;
594  BEGIN
595      select Balance into v_balance
596      from Accounts
597      where AccountId=:NEW.AccountID;
598      IF :NEW.TransactionType='Deposit' and :NEW.Amount<=0 THEN
599          RAISE_APPLICATION_ERROR(-20003,'Deposit amount must be positive');
600      end if;
---
```

Query result

Script output

DBMS output

Explain Plan

SQL history



ORA-20003: Deposit amount must be positive
ORA-06512: at "AKSHAYABALABHADRA_SCHEMA_BA5N8.CHECKTRANSACTIONRULES", line 8
ORA-04088: error during execution of trigger 'AKSHAYABALABHADRA_SCHEMA_BA5N8.CHECKTRANSACTIONRULES'

<https://docs.oracle.com/error-help/db/ora-20003/>
Error at Line: 4 Column: 0

SQL> INSERT INTO Transactions(TransactionID,AccountID,TransactionDate,Amount,TransactionType)
VALUES (9,1,SYSDATE,999999,'Withdrawal')

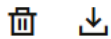
ORA-20004: Withdrawal amount exceeds account balance
ORA-06512: at "AKSHAYABALABHADRA_SCHEMA_BA5N8.CHECKTRANSACTIONRULES", line 14
ORA-04088: error during execution of trigger 'AKSHAYABALABHADRA_SCHEMA_BA5N8.CHECKTRANSACTIONRULES'

<https://docs.oracle.com/error-help/db/ora-20004/>

[SQL Worksheet]*      Aa 

```
615 --Cursor: Scenario-1
616
617 set SERVEROUTPUT on;
618 DECLARE
619     ...cursor c_mon_statements IS
620     ...select c.Name,t.TransactionID,t.Amount,t.TransactionType,t.TransactionDate
621     ...from TRANSACTIONS t
622     ...join Accounts a on t.AccountID=a.AccountId
623     ...
```

Query result **Script output** DBMS output Explain Plan SQL history



Statement for: Amit Verma
TxnID: 10 | Deposit | Amount: 7000 | Date: 04-JUN-2026
TxnID: 3 | Deposit | Amount: 3000 | Date: 17-JUN-2026
Statement for: Kiran Kumar
TxnID: 5 | Deposit | Amount: 2500 | Date: 12-JUN-2026
Statement for: Neha Joshi
TxnID: 8 | Withdrawal | Amount: 500 | Date: 11-JUN-2026
Statement for: Priya Reddy
TxnID: 2 | Withdrawal | Amount: 2000 | Date: 16-JUN-2026
Statement for: Rahul Sharma
TxnID: 9 | Withdrawal | Amount: 1000 | Date: 07-JUN-2026
TxnID: 1 | Deposit | Amount: 5000 | Date: 14-JUN-2026
TxnID: 13 | Deposit | Amount: 150 | Date: 19-JUN-2026
Statement for: Sneha Patel
TxnID: 4 | Withdrawal | Amount: 1000 | Date: 09-JUN-2026
Statement for: Sneha Patel Updated
TxnID: 7 | Deposit | Amount: 3500 | Date: 18-JUN-2026

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.014

[SQL Worksheet]*



Aa



```
641
642  --scenario-2
643
644  set SERVEROUTPUT on;
645  declare
646      ... cursor c_accounts is
647          ... select AccountID, Balance
648          ... from Accounts
649          ... for update;
```

Query result

Script output

DBMS output

Explain Plan

SC



Elapsed: 00:00:00.014

```
SQL> declare
      cursor c_accounts is
      select AccountID, Balance
      from Accounts...
```

[Show more...](#)

Account 1: Fee of 50 deducted. New balance: 14494.98
Account 2: Fee of 50 deducted. New balance: 8600
Account 3: Fee of 50 deducted. New balance: 25402
Account 4: Fee of 50 deducted. New balance: 4900
Account 5: Fee of 50 deducted. New balance: 12140.7
Account 7: Fee of 50 deducted. New balance: 18261.3
Account 8: Fee of 50 deducted. New balance: 2900

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.007

[SQL Worksheet]*



Aa



```
664  --scenario-3
665
666  set SERVEROUTPUT on;
667  declare
668      cursor c_loans is
669          select LoanID, LoanAmount, INTERESTRATE
670          from loans
671          for update;
672      v_new_policy_rate CONSTANT number:=7;
673  BEGIN
674      for loan_rec in c_loans LOOP
```

Query result

Script output

DBMS output

Explain Plan

SQL history



```
SQL> declare
      cursor c_loans is
          select LoanID, LoanAmount, INTERESTRATE
          from loans...
```

[Show more...](#)

Loan 1: Rate changed from 5.67% to 7%
Loan 2: Rate changed from 7% to 7%
Loan 3: Rate changed from 5.67% to 7%
Loan 4: Rate changed from 7% to 7%
Loan 5: Rate changed from 5.67% to 7%
Loan 6: Rate changed from 7% to 7%
Loan 7: Rate changed from 7% to 7%
Loan 8: Rate changed from 5.67% to 7%

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.008

[SQL Worksheet]*



Aa



```
685
686  --Packages: Scenario-1
687
688  CREATE OR REPLACE PACKAGE CustomerManagement
689  AS
690      ... PROCEDURE AddCustomer
691      ... (
692          ... p_customer_id IN NUMBER,
693          ... p_name IN VARCHAR2,
694          ... p_dob IN DATE,
695          ... p_balance IN NUMBER
696      ... );
697      ... PROCEDURE UpdateCustomerDetails (
698          ... p_customer_id IN NUMBER,
699          ... p_name IN VARCHAR2,
700          ... p_balance IN NUMBER
701      ... );
702      ... FUNCTION GetCustomerBalance (p_customer_id IN NUMBER) RETURN NUMBER;
703  END CustomerManagement;
704  /
705  CREATE OR REPLACE PACKAGE BODY CustomerManagement
```

Query result

Script output

DBMS output

Explain Plan

SQL history



ERROR: Customer ID already exists.
Customer 7 updated.
Balance: 3500

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.012

[SQL Worksheet]*



Aa



```
761
762  --scenario-2
763
764  CREATE OR REPLACE PACKAGE EmployeeManagement
765  AS
766      ... PROCEDURE HireEmployee
767      ... (
768          ... p_employee_id IN NUMBER,
769          ... p_name IN VARCHAR2,
770          ... p_position IN VARCHAR2,
771          ... p_salary IN NUMBER,
772          ... p_department IN VARCHAR2
773      ... );
774      ... PROCEDURE UpdateEmployeeDetails
775      ... (
776          ... p_employee_id IN NUMBER,
777          ... p_position IN VARCHAR2,
778          ... p_salary IN NUMBER
779      ... );
780      ... FUNCTION CalculateAnnualSalary(p_employee_id IN NUMBER) RETURN NUMBER;
781  END EmployeeManagement;
```

Query result

Script output

DBMS output

Explain Plan

SQL history



ERROR: Employee ID already exists.
Employee 4 updated.
Annual Salary: 720000

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.019

[SQL Worksheet]*



Aa



```
841
842  --scenario-3
843
844  create or replace PACKAGE AccountOperation
845  AS
846    ... PROCEDURE OpenAccount
847    ... (
848    ...    p_account_id in number,
849    ...    p_customer_id in number,
850    ...    p_account_type in VARCHAR2,
851    ...    p_balance in number
852    ... );
853    ... PROCEDURE CloseAccount(p_account_id in number);
854    ... FUNCTION GetTotalBalance(p_customer_id in number) return number;
855  end AccountOperation;
856  /
857  create or replace PACKAGE BODY AccountOperation
858  AS
859    ... PROCEDURE OpenAccount
860    ... (
861    ...    p_account_id in number,
```

Query result

Script output

DBMS output

Explain Plan

SQL history



Account 6 opened for Customer 1
Total Balnce for Customer 1:16494.98
Account 6 closed.

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.017